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Basics of data-driven Climate Smart Agriculture

Module 2.3.2

Data and analyses at the Farm

Nappaa hiilestä kiinni
MAANKÄYTTÖSEKTORIN ILMASTORATKAISUT

Nature of Data

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An over-simplified informal way of answering the question ”What is data?” could be

A symbolic description of what can be known

Letters, numbers, etc

Text notes, signal, numeric records, images etc

Temperature, Weight

Measured, observed

Data value chain

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Consider each term below and ask yourself – what is it made up of and how does one create it?

Knowledge

Information

Wisdom

Data

Data value chain

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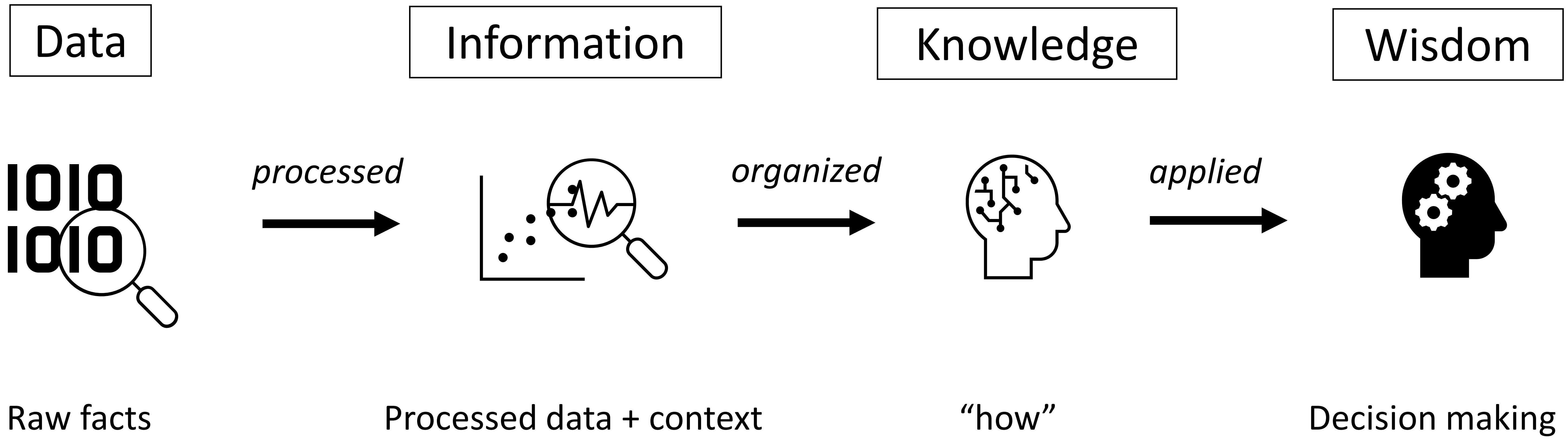

Commonly referred to as the DIKW hierarchy where each node creates value consolidating and assessing content from the previous node(s) and can be used to answer increasingly complex questions

DIKW hierarchy – farm specific examples

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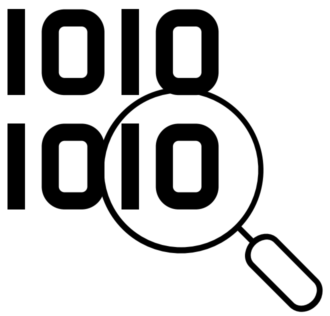
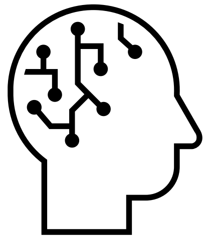
Data		Information		Knowledge		Wisdom
	*processed* →		*organized* →		*applied* →	
Raw facts		Processed data + context		"how"		Decision making
Temperature *Soil moisture* *Yield* …		*Yield distribution* *Vegetation indices* *Growing degree days* …		*Yield gap* *Seed trials* *Soil health* …		*Cropping system* *Irrigation scheduling* …

Agridata

It is common to hear the refrain – “Age of (*big*) data” – where new and promising data analysis applications and results in agricultural sectors are constantly being published.

Thought exercise: Is appreciation of data in farming new?

Agridata

The Sumerians, considered to have invented the first accounting and writing systems, kept record of harvest and grain inventories. The clay cuneiform tablet (right) from ca. 3100–2900 BCE most likely documents the delivery and distribution of barley and emmer wheat.

https://www.metmuseum.org/art/collection/search/327384

Agridata

Clearly, appreciation for agridata is not new.

The essential challenge, however, is to use advancements in sensor, communication and analysis technology to explore advanced applications and knowledge discovery.

With greater understanding of the ecosystem interdependencies, data can be used to not only increase productivity, but importantly to do it sustainably.

https://www.serg.ai/2021/03/the-long-history-of-data-in-agriculture/

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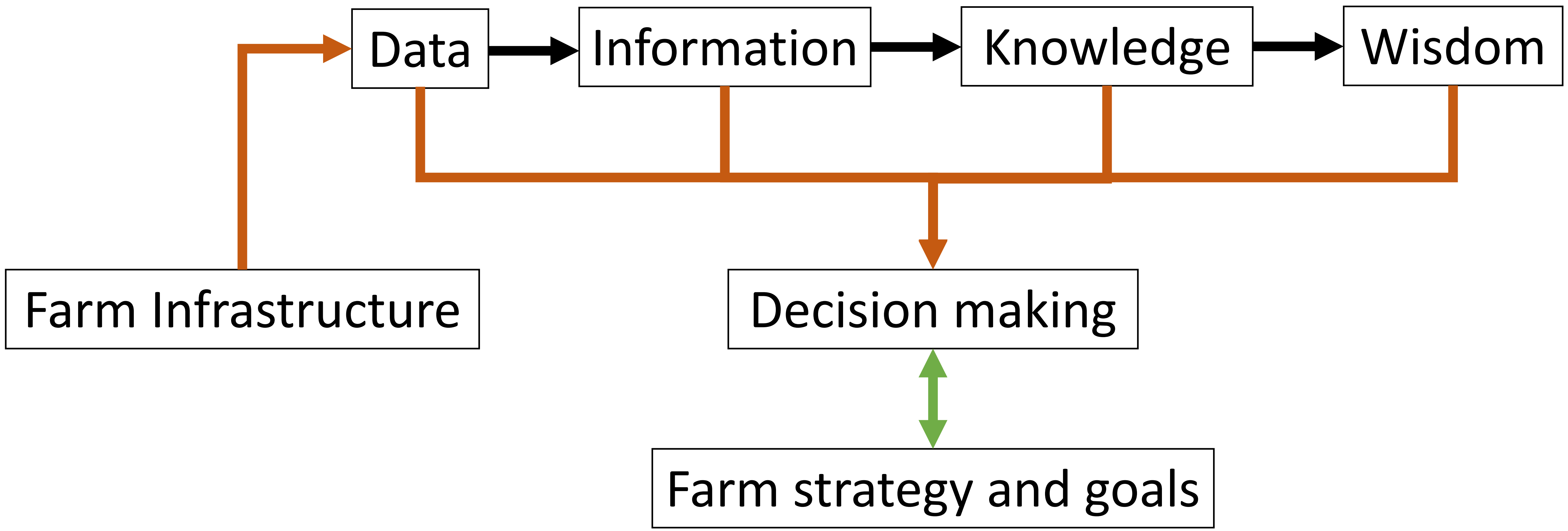
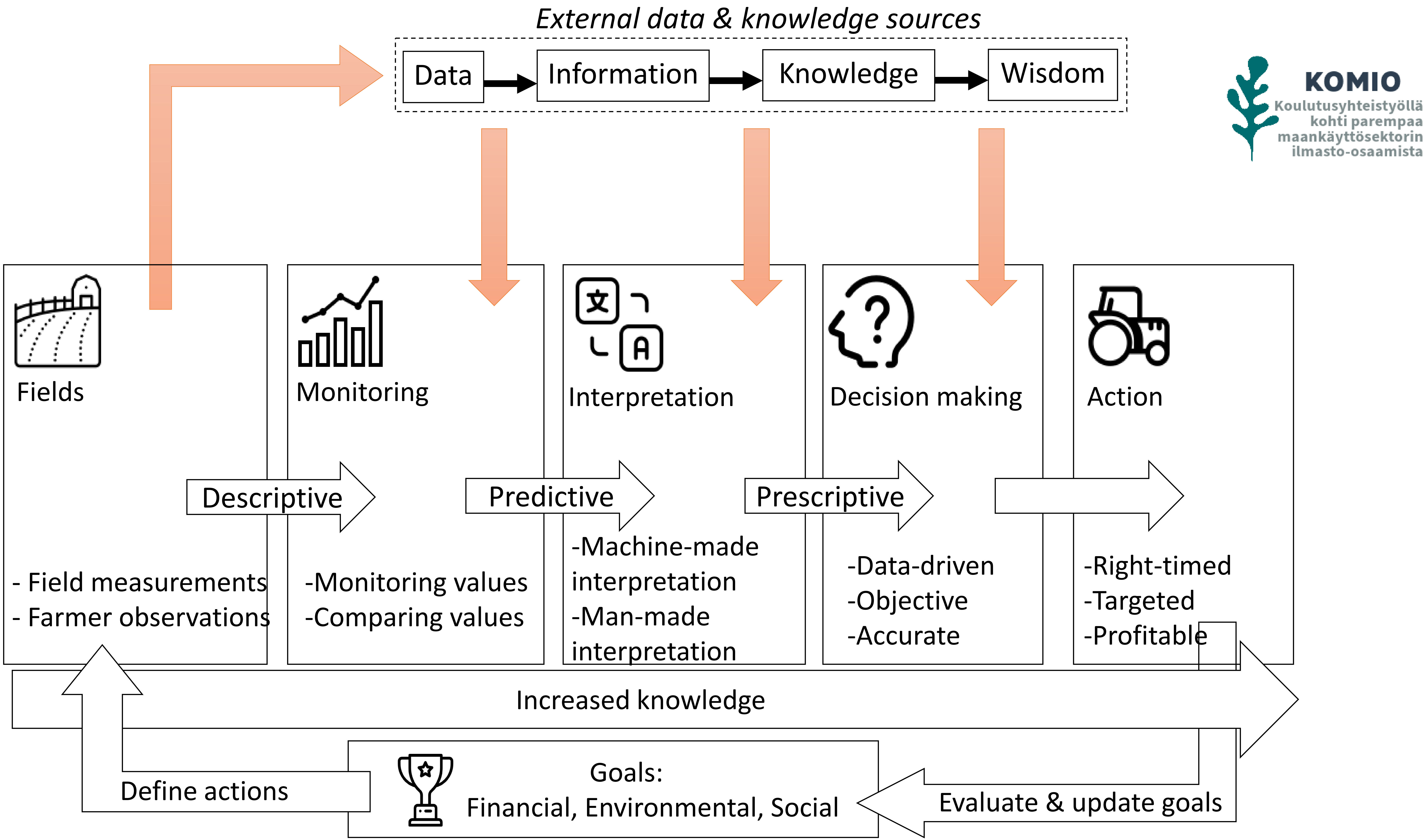
Data-driven on-farm decision making

The data value chain positions itself by helping decision making within the context of farm-specific strategy and goals.




External data & knowledge sources
Data
Information
Knowledge
Wisdom
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Fields
Monitoring
Interpretation
Decision making
Action
Descriptive
Predictive
Prescriptive
- Field measurements
- Farmer observations
-Monitoring values
-Comparing values
-Machine-made interpretation
-Man-made interpretation
-Data-driven
-Objective
-Accurate
-Right-timed
-Targeted
-Profitable
Increased knowledge
Define actions
Goals:
Financial, Environmental, Social
Evaluate & update goals

Sources of agridata

Examples of data categorized according to source locations:

<u>On-farm data</u>

- Notes
- Sensors
- IoT devices
- Weather

<u>National data</u>

- Weather (e.g. FMI)
- Soil maps

<u>International data</u>

- Remote sensing
- Market condition monitoring
- Sample surveys

Data streams: CSA context

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For any farm, data sources can be identified by mapping the data categories and sub-categories by starting from strategy and goals set within the CSA context.

A map of such data streams can help with capturing the overall layout of what data are currently available either on-farm or accessible via web platforms.

Data streams: CSA context

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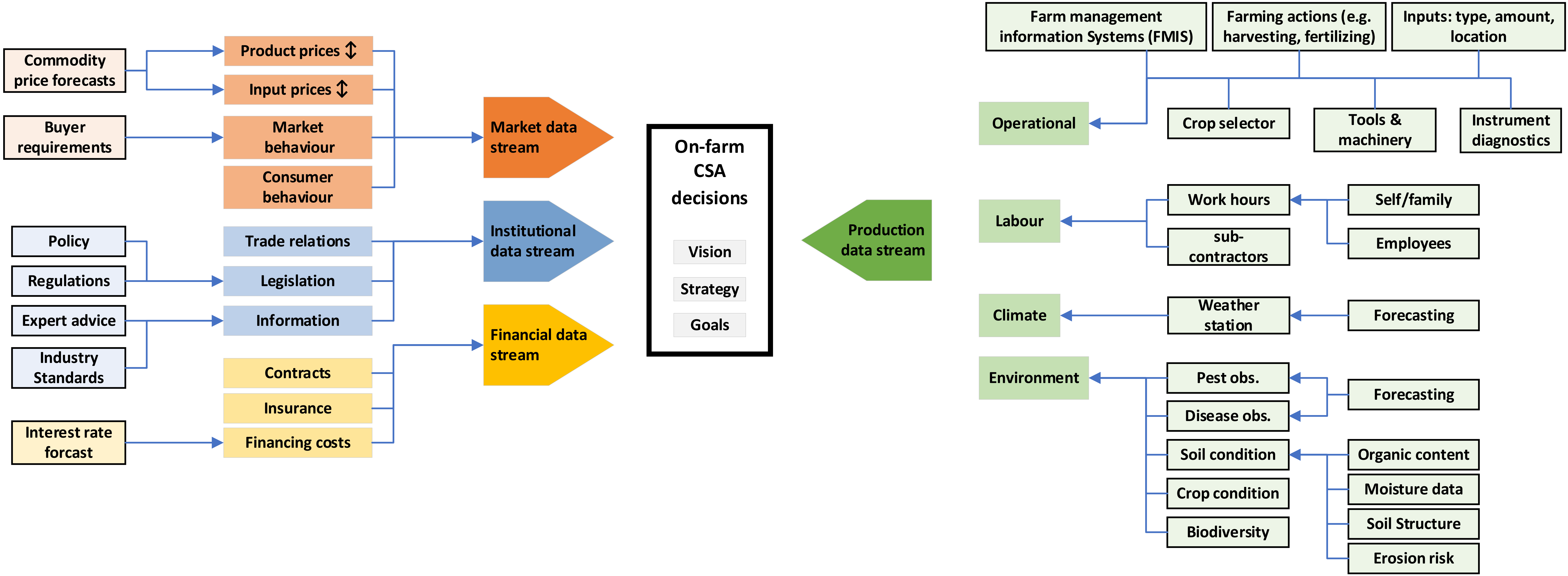

Thought exercise:

Consider how each source is stored and accessed –

- Method of storage
- Method of retrieval
- Mode of access

Self-study exercises

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In the following slides 3 cases of data portals, whose location is referenced, the reader can explore how agridata is collected and displayed

Case: Field Observatory

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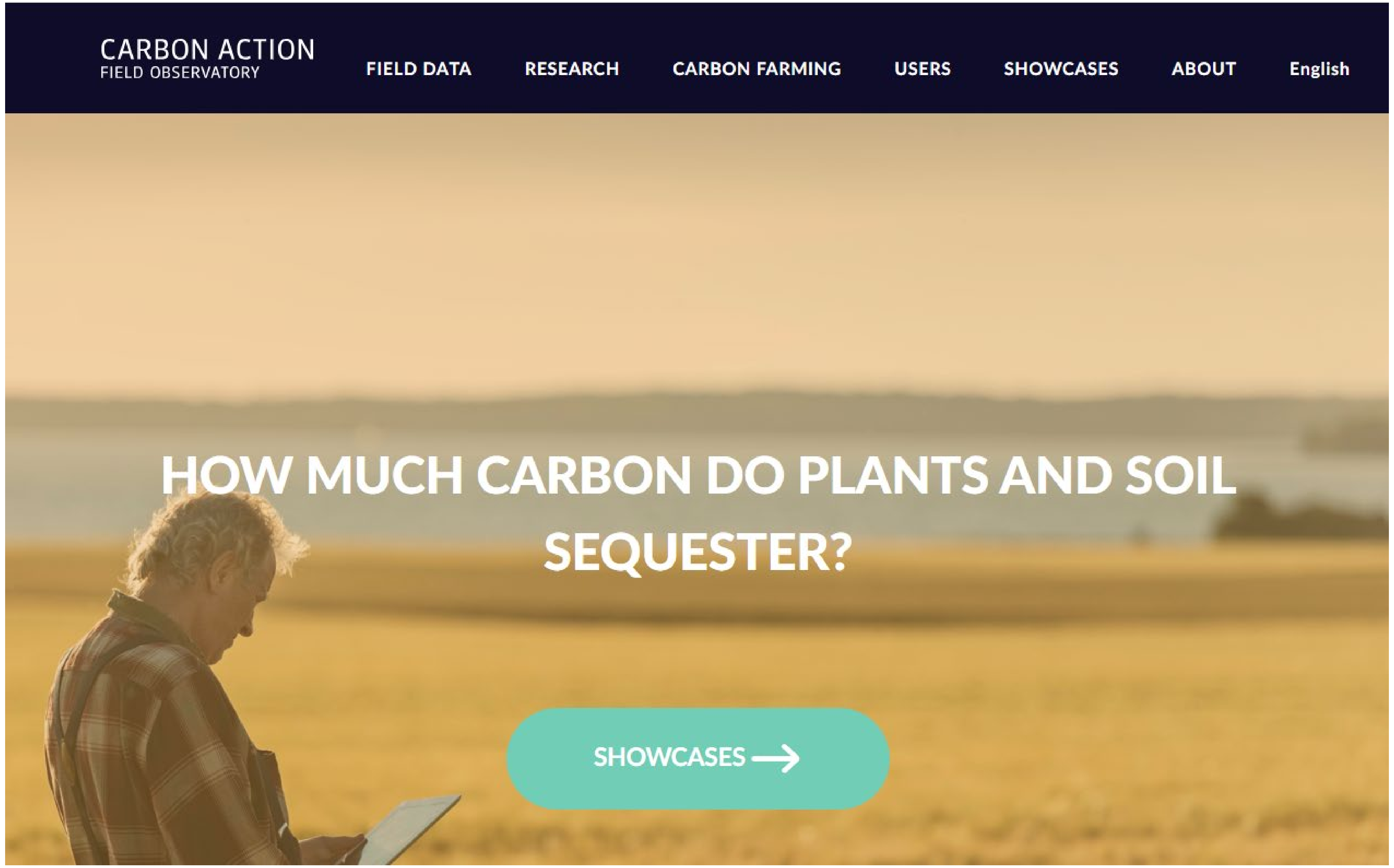
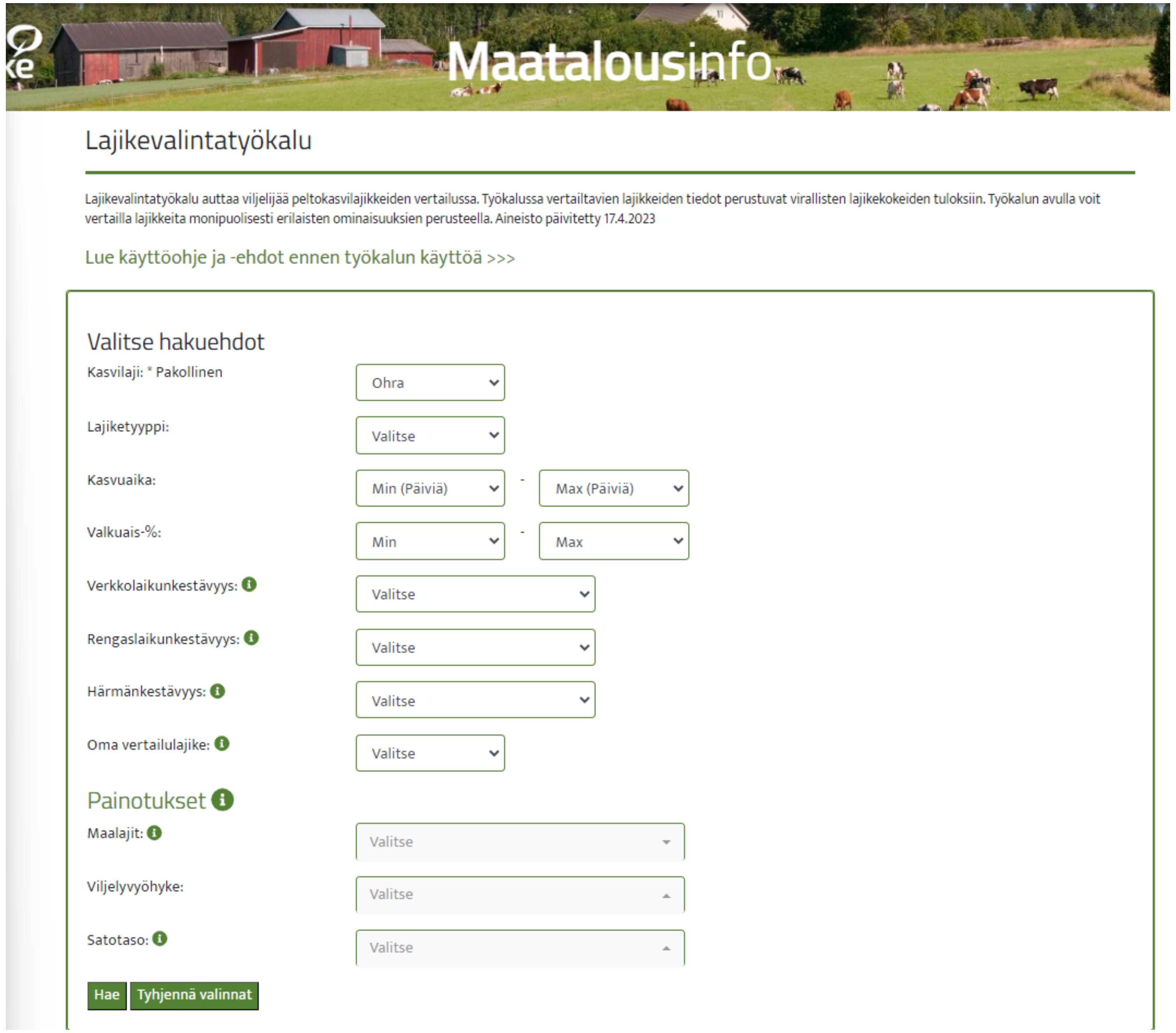


https://www.fieldobservatory.org/index.php/home/

Case: Crop variety selector



https://maatalousinfo.luke.fi/fi/lajikevalinta

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Case Market condition

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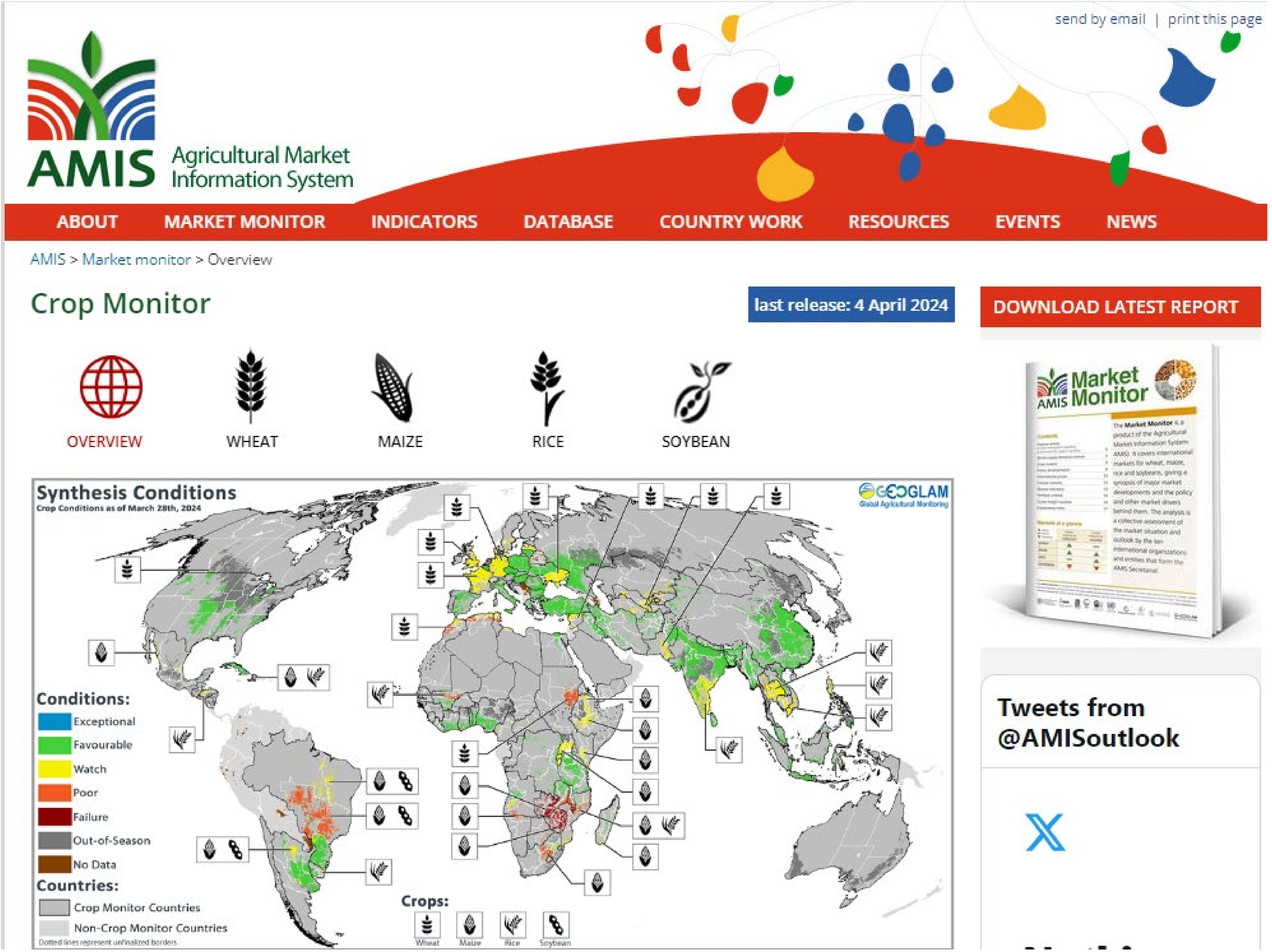

https://www.amis-outlook.org/amis-monitoring/monthly-report/en/

Value of data is in the knowledge it fosters

Thought exercise: what would be some examples of useless data? Is there such a thing?

Agridata analysis

Different steps in the data value chain require varying levels of data analysis methods to extract the desired parameters or explore unknown relationships.

The analyses can be broadly categorized as:

- Descriptive – exhibits the state of parameters in a system/subsystems in the farm ecosystem. Examples: soil nutrient content, soil moisture, plant biomass
- Predictive – The state of any parameter can forecast based on current information and knowledge of a systems response. Examples: weather, yield prediction
- Prescriptive – the state of a system can be manipulated/adjusted using current data and forecasting effects of interventions. Examples: Variable rate application of soil amendments, Irrigation scheduling, cropping system planning

Analysis process

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A typical analysis is executed through a sequence of steps as illustrated below

Define Purpose → Obtain data → Explore & clean data → Identify analysis task → Choose analysis methods → Technical implementation & optimization

Analysis process

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The process is initiated by defining the objective of the analysis – i.e., identifying an 'unknown' in a farming operation that is important for decision making

Define Purpose → Obtain data → Explore & clean data → Identify analysis task → Choose analysis methods → Technical implementation & optimization

Analysis process

Some examples of such a purpose:

- How much feed does a cow consume in a certain time period at a specific parcel and how does this relate to the milk production in that period?
- How to exploit big data on earth observation (EO) systems and datasets?
- How to characterize soil and plants effectively?
- How to perform dairy herd culling more productively?
- How to assure the safety and the quality of feedstuffs for animals?
- How to protect small farmers against droughts?
- How to consider the opportunities of using drought-tolerant maize varieties in Southern Africa?
- How to forecast weather changes?
- How to classify and profile soil?
- How to identify management zones for cotton production?
- How to classify land use and land cover changes, as well as map crops?
- How to create accurate inventories of grasslands using remote sensing?
- How to determine phenological stages of paddy rice?
- How to support food chain safety measures for cattle holdings?
- How to identify weeds in sunflower crops to minimize the impact of herbicide?
- How to find spatial equilibrium and optimal locations of agricultural facilities?
- How to improve the productivity and income of smallholder farmers?
- How to properly identify crops to study crop rotation?
- How to evaluate fish and wildlife population viability under land management alternatives?
- How to better understand how management choices affect sustainability performance and operational efficiency?
- How to recognize animal diseases?
- How to estimate food availability in sub-Saharan Africa?
- How to facilitate farmers' financing and easier payments?
- How to provide fair and enticing insurance and finance for farmers?

Analysis process

Having defined a purpose, all appropriate data need to be identified and aspects of data collection and storage need to be addressed.

Define Purpose → **Obtain data** → Explore & clean data → Identify analysis task → Choose analysis methods → Technical implementation & optimization

- *Data collection plan?*
- *How is it to be stored?*
- *Data ownership and privacy?*

Case example: A producer may wish to map out all the data sources relevant and/or available to their operation. Each data type may have its own protocol for access and storage

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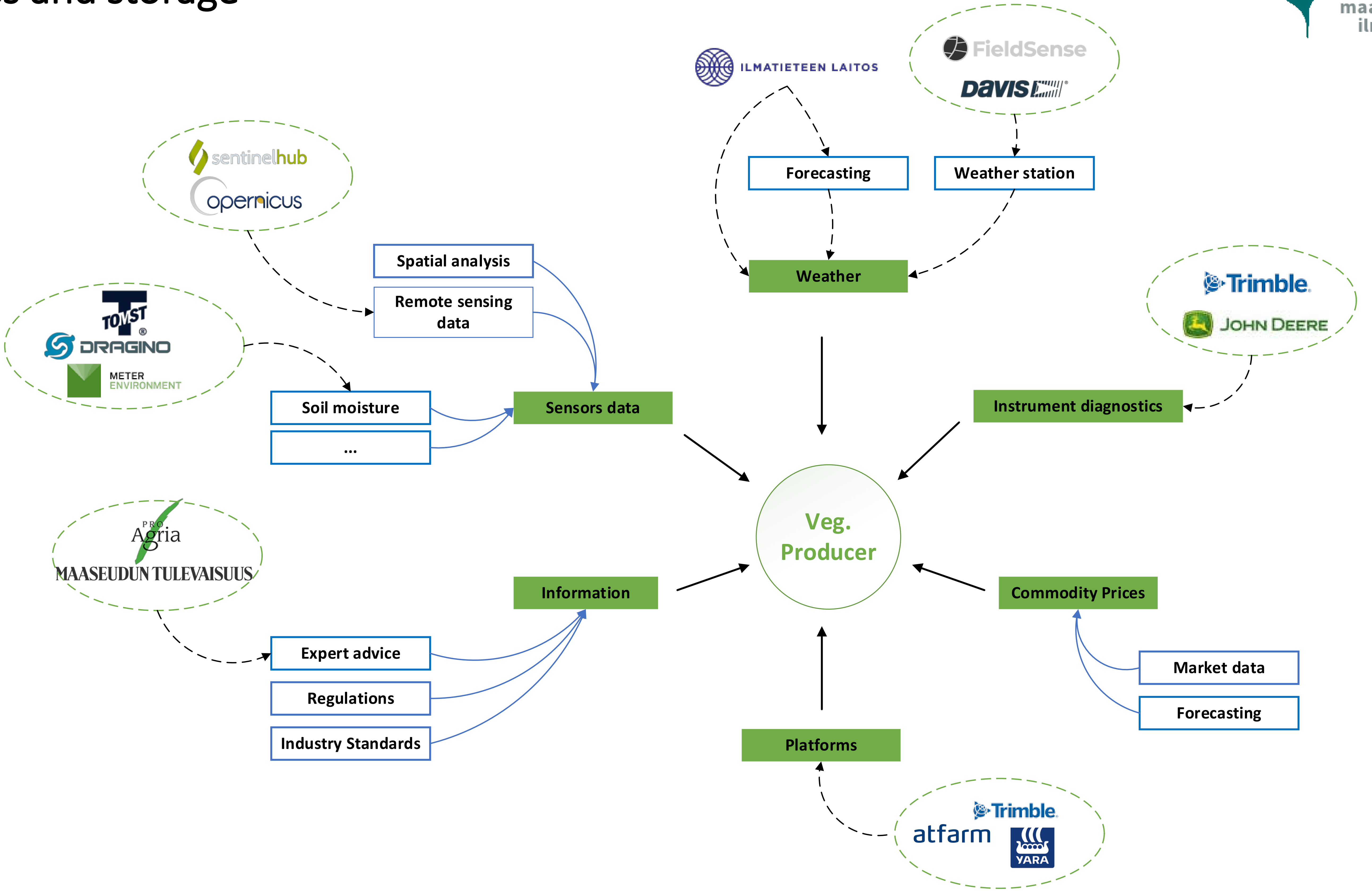

Analysis process

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Corrupt or missing data, often present, needs to be cleaned. Data coherence may need to be confirmed.

Define Purpose → Obtain data → **Explore & clean data** → Identify analysis task → Choose analysis methods → Technical implementation & optimization

- *Data consistency issues?*
- *Missing data?*
- *Collection strategy deficiencies?*
- *Nature of data – e.g., distribution?*
- *Units and calibration?*
- *Derived parameters?*

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Visualization helps with understanding the nature of collected data and exploring cursory relationships that can help identifying appropriate methods downstream.

Analysis process

Define Purpose → Obtain data → Explore & clean data → **Identify analysis task** → **Choose analysis methods** → Technical implementation & optimization

With a sound understanding of the acquired data and required type of analysis (descriptive, predictive or prescriptive), the appropriate task(s) is identified, and method(s) chosen.

E.g. if the task is that of classification, then one or multiple classification methods can be chosen to test on the data

Analysis process

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Define Purpose → Obtain data → Explore & clean data → **Identify analysis task** → **Choose analysis methods** → Technical implementation & optimization

Some method examples from literature used in data analysis:

- Machine learning
- Cloud platforms
- Wavelet analysis
- Spectral data analysis
- Decision support systems
- Geospatial analysis
- Mobile applications
- Modelling & simulation
- Statistical analysis
- Web services
- Computer vision
- Fourier transform
- Image processing

Analysis process

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Define Purpose → Obtain data → Explore & clean data → Identify analysis task → Choose analysis methods → Technical implementation & optimization

Numerous sophisticated methods are grouped under “Artificial Intelligence”. Among these, Machine learning methods have in the past couple of decades made tremendous strides in capability in handling big data and learning complex relationships buried in the data.

Sensor Data → Artificial Intelligence Methods → Agronomy

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Temperature
Solar radiation
Precipitation
Humidity
…

Irrigation
Fertilizers
Compost
Herbicides
…

+ Remote sensing image data

Soil type
Mineral content (N,P,K,..)
Organic content
Moisture

Machine Learning

Deep Learning

CNN (Convolutional Neural Networks)

Agriculture information processing

Agriculture production system optimal control

Smart agriculture machinery equipment

Agricultural economic system management

Sensor Data → **Artificial Intelligence Methods** → **Subject Areas**

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Temperature
Solar radiation
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Herbicides
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Machine Learning

Deep Learning

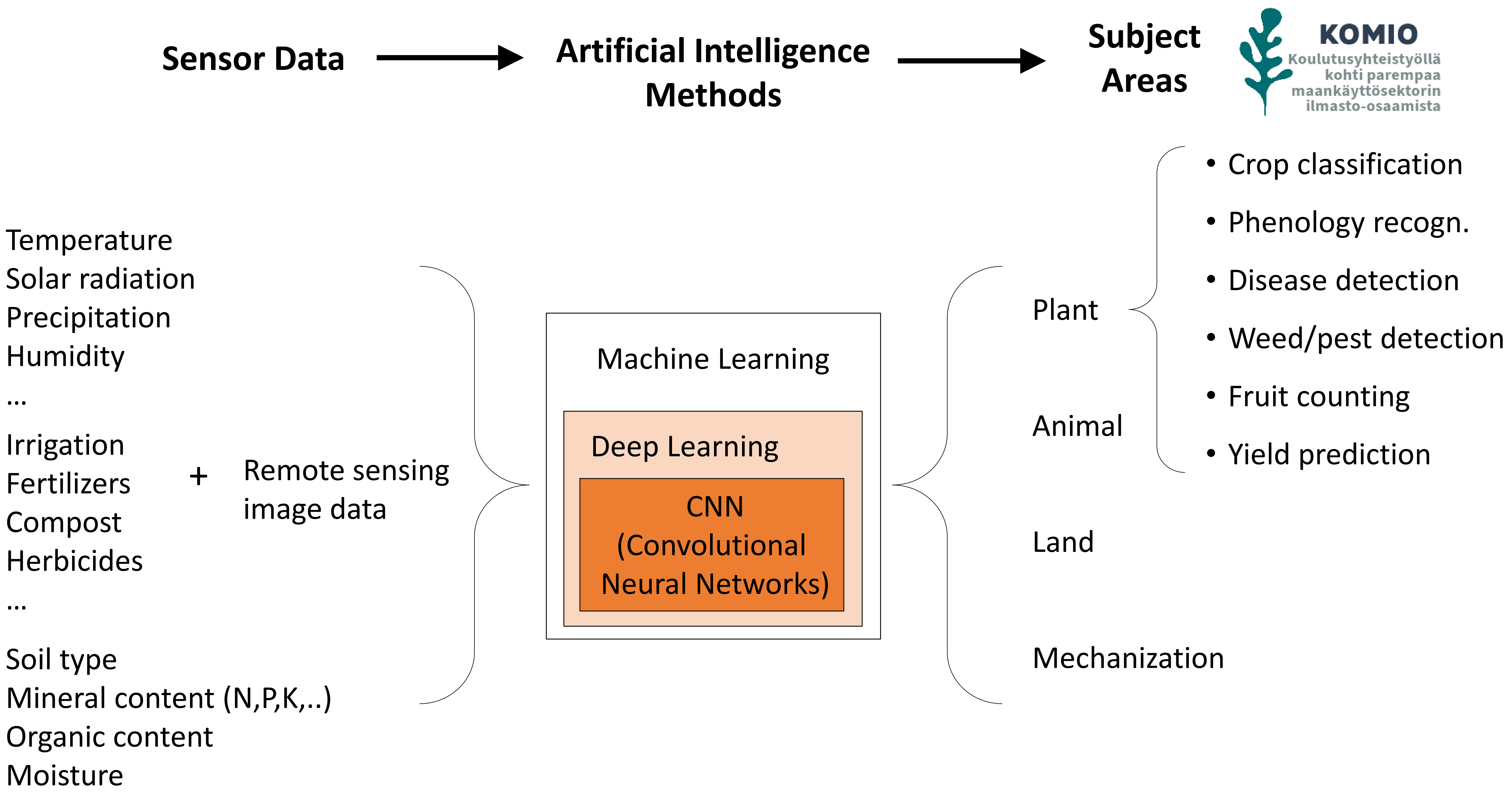
CNN (Convolutional Neural Networks)

Plant

Animal

Land

Mechanization



Sensor Data
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Plant
Animal
Land
Mechanization
Crop classification
Phenology recogn.
Disease detection
Weed/pest detection
Fruit counting
Yield prediction

Machine learning applications

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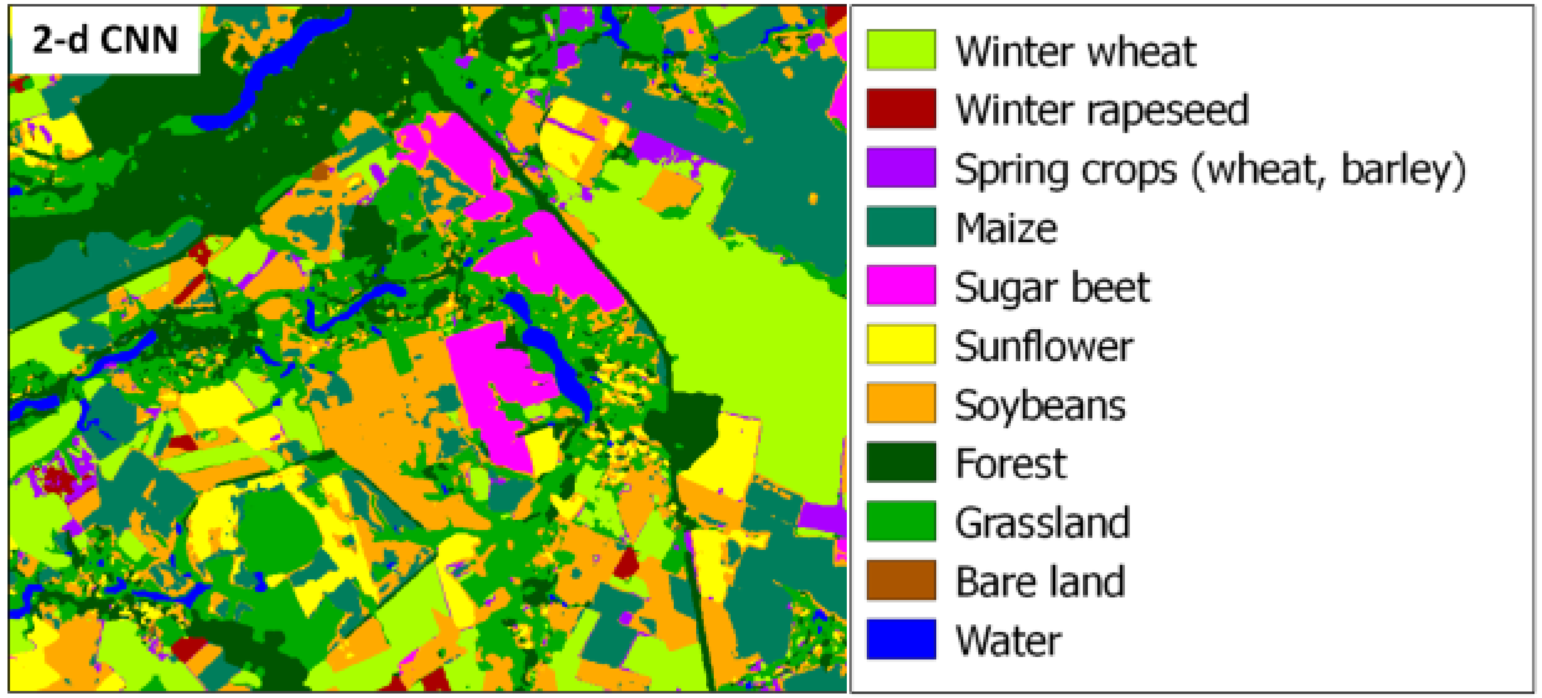
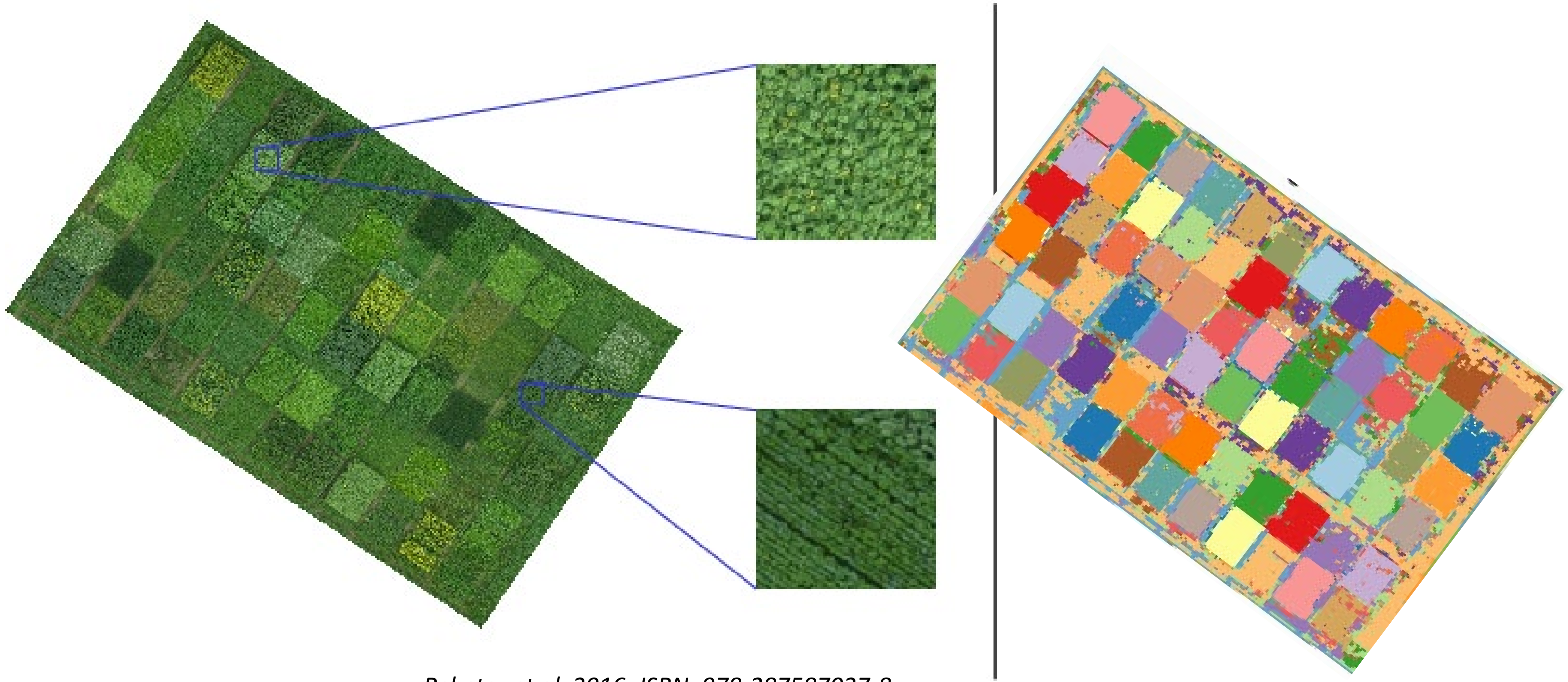
- The following slides show examples from academic literature of applications from each of the subject areas
- The reader is encouraged to read the referenced publications to understand how the applications were implemented



Kussul et al. 2017; DOI: 10.1109/JSTARS.2016.2560141



Rebetez et al. 2016; ISBN: 978-287587027-8

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- **Crop classification**
- Phenology recogn.
- Disease detection
- Weed/pest detection
- Fruit counting
- Yield prediction

Dataset	Method	Precision	Recall	F1-Score	Accuracy
		(%)	(%)	(%)	(%)
Cotton	CNN-BA	87.32	86.14	86.58	86.54
Pepper	CNN-BA	88.12	87.24	87.28	87.14
Corn	CNN-BA	87.32	86.14	86.58	86.54

Yalcin, Hulya. "Plant phenology recognition using deep learning: Deep-Pheno." 2017 6th International Conference on Agro-Geoinformatics (2017): 1-5.

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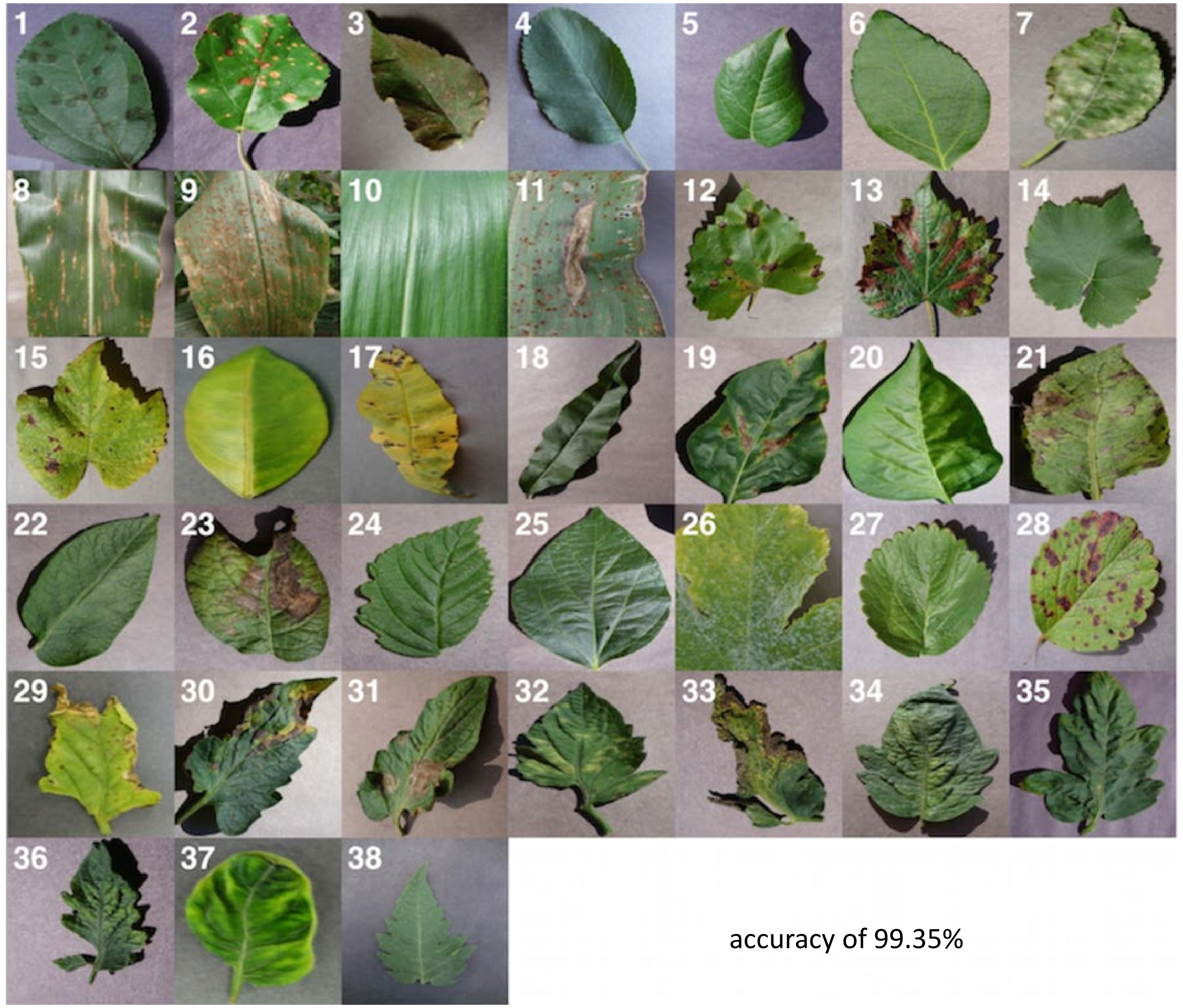

accuracy of 99.35%

Mohanty et al. 2016; DOI: 10.3389/fpls.2016.01419

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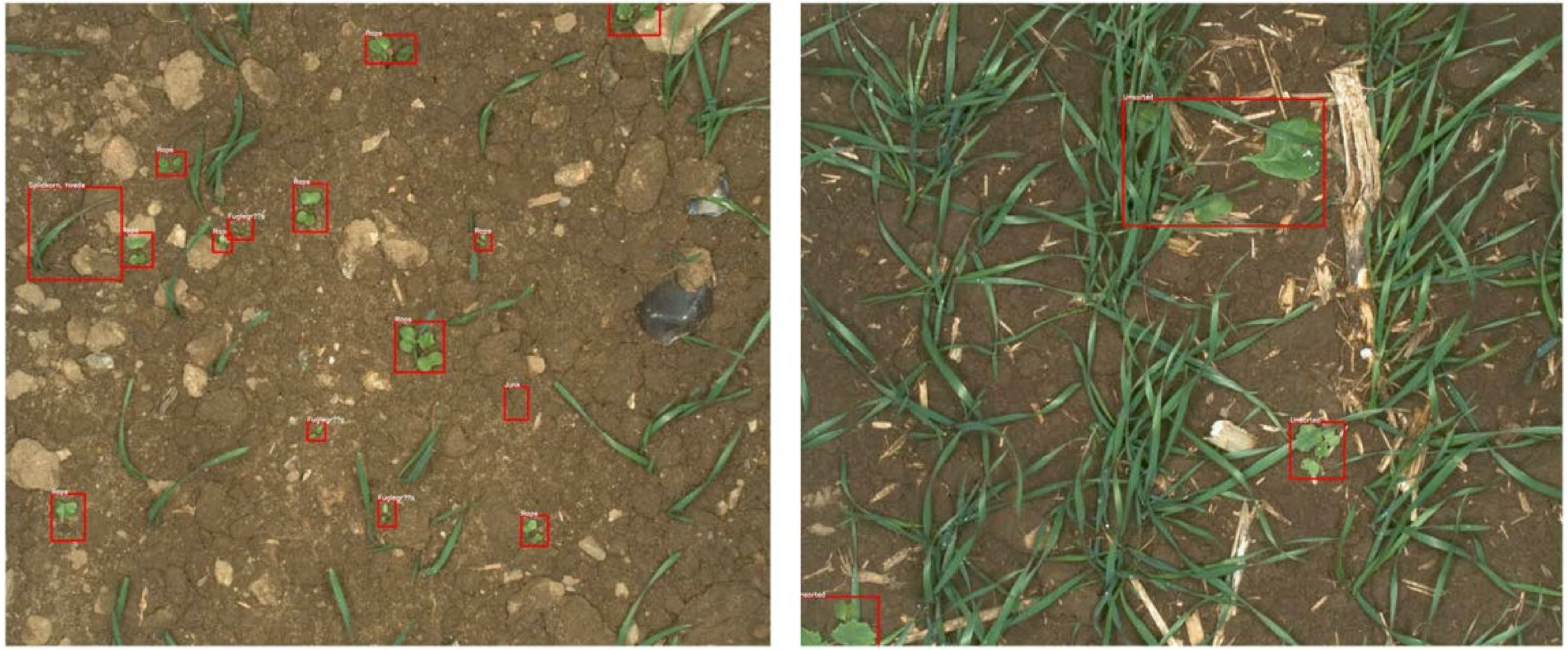
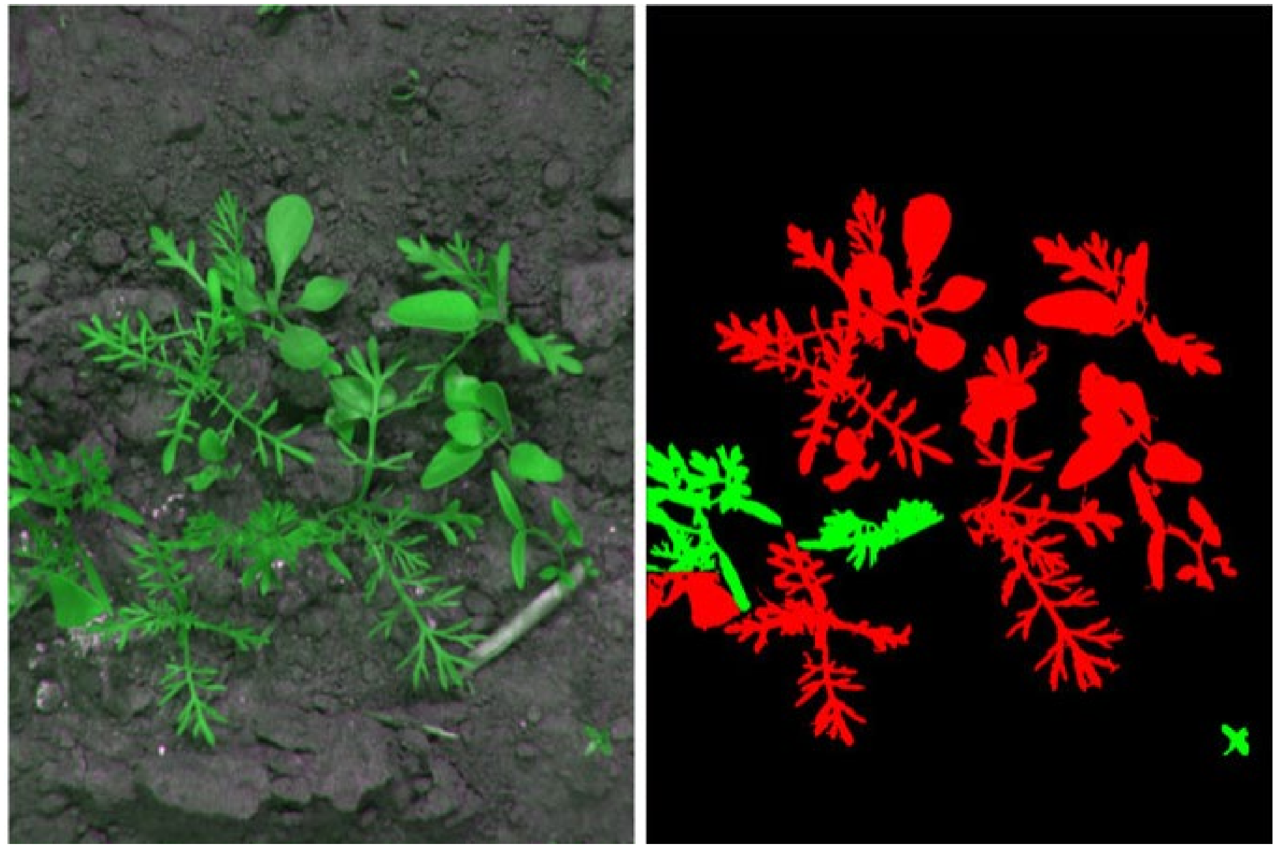


Dyrmann et al. 2017; DOI: 10.1017/S2040470017000206



McCool et al. 2017; DOI: 10.1109/LRA.2017.2667039

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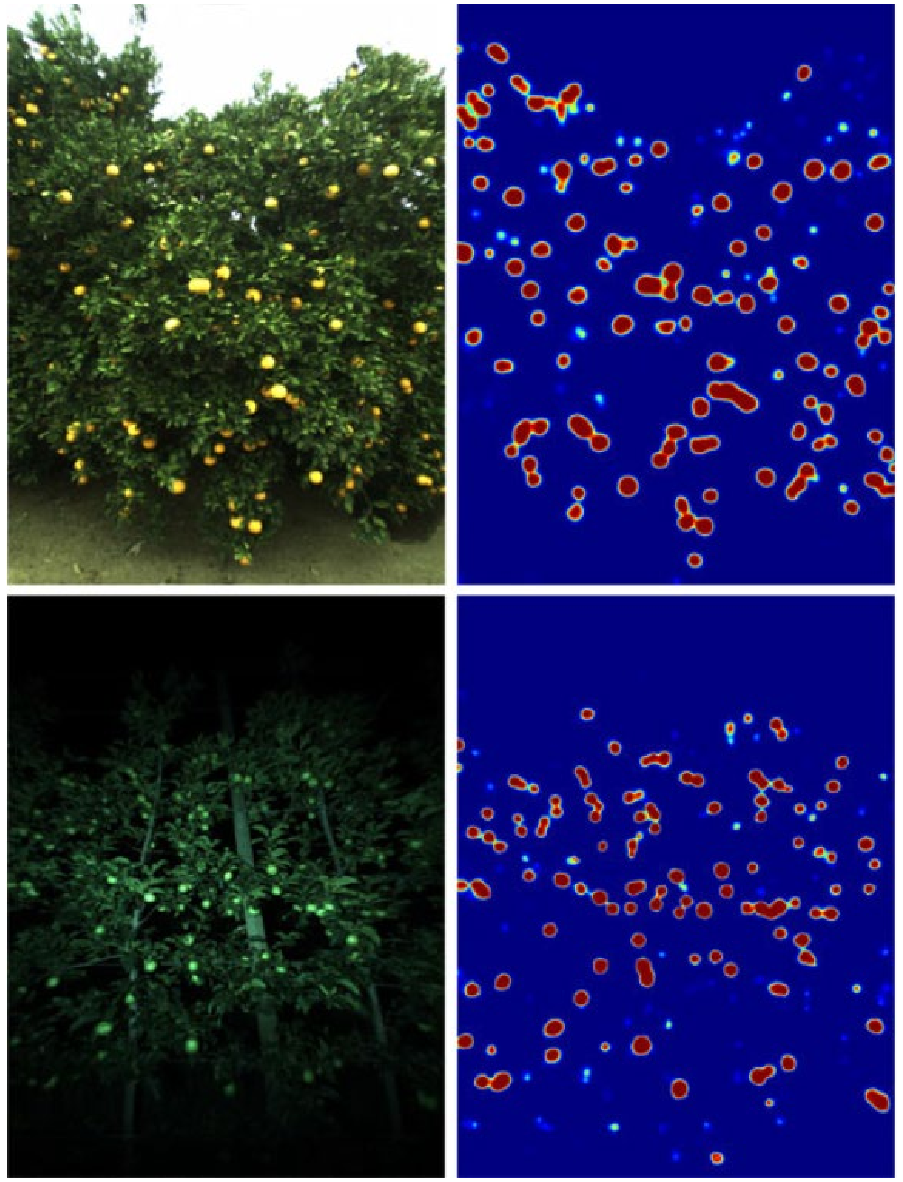
Chen et al. 2017; DOI: 10.1109/LRA.2017.2651944



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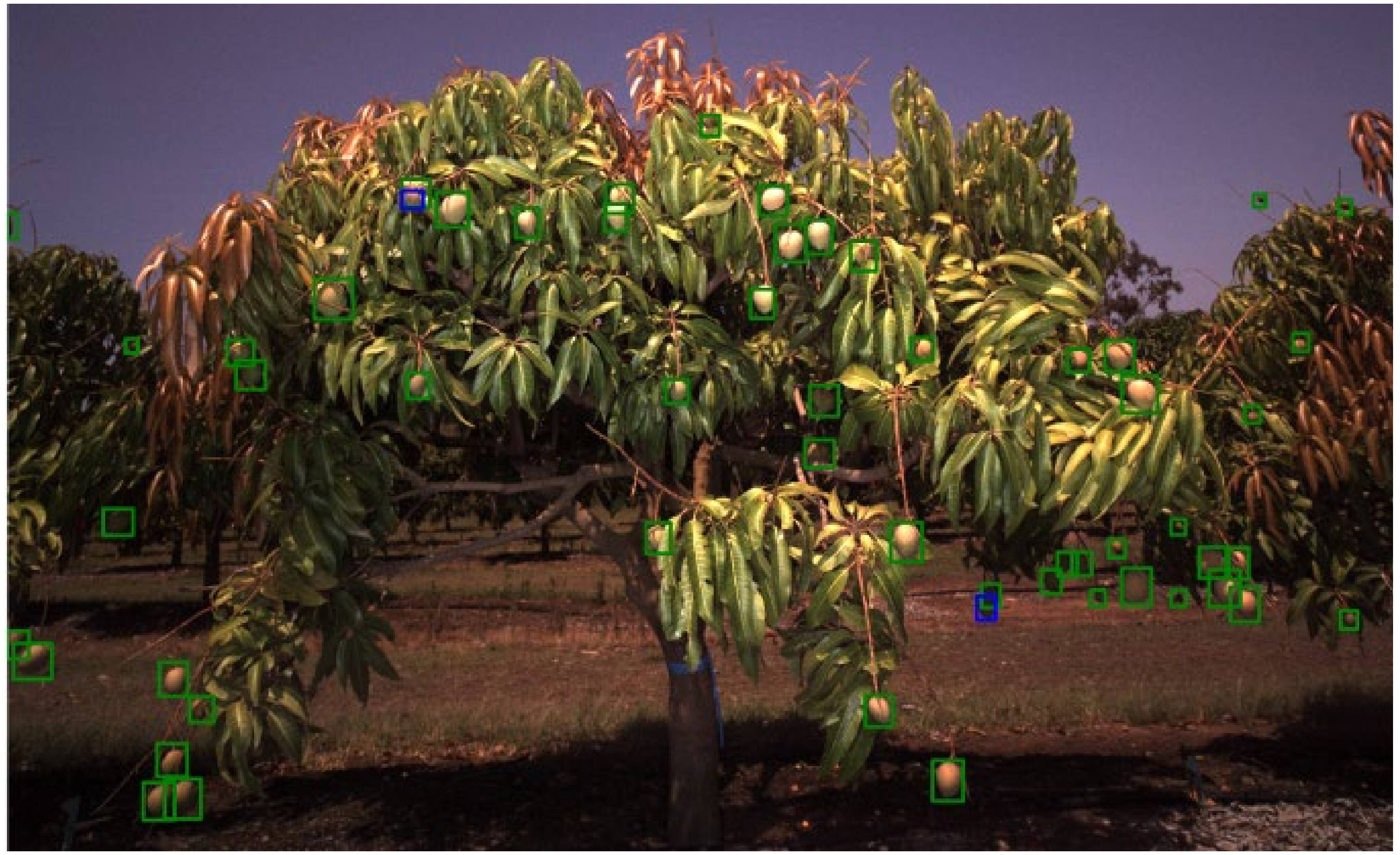
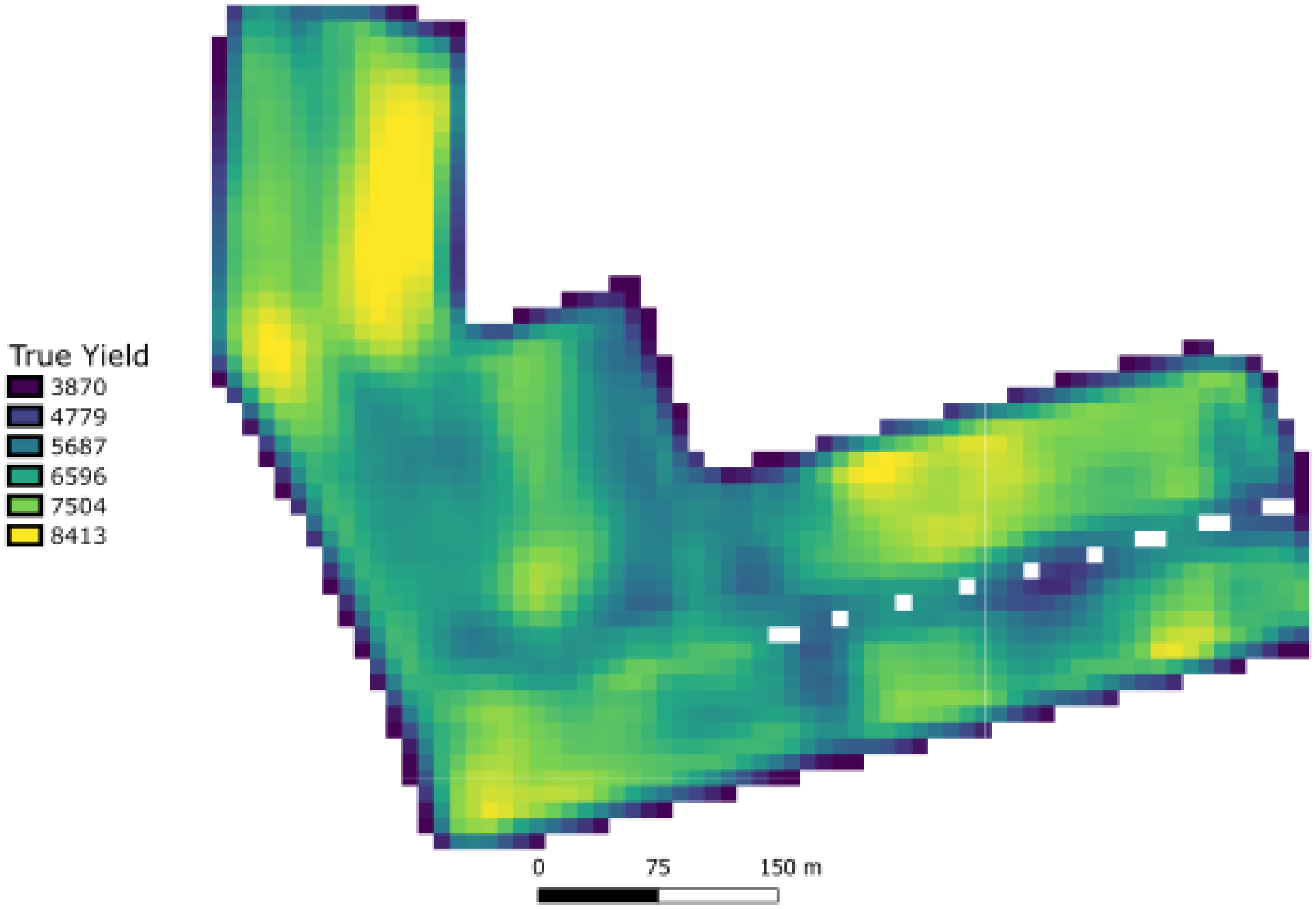
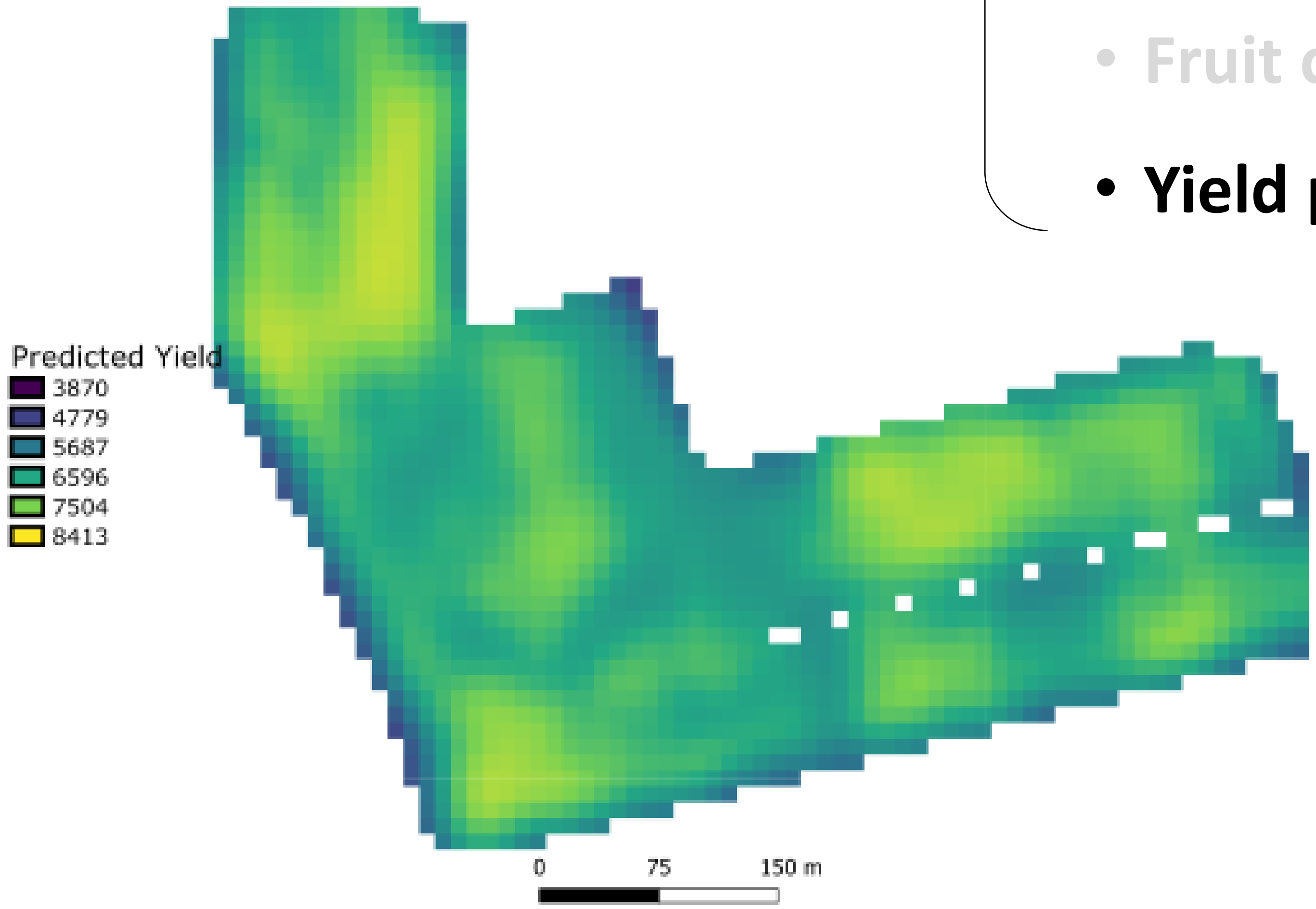


Bargoti & Underwood 2016; arXiv:1610.03677v2

Yield prediction





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